



**B.Sc. (General/Special) Degree
Programme
Faculty of Applied Sciences
University of Sri Jayewardenepura**

ICT 1022 2.0 Computer System Architecture

Presented By:

Surani Tissera(PhD)

Department of Computer Science

Basic Operation of a Computer



This is a property of Department of Computer Science, Faculty of Applied Science, University of Sri Jayewardenepura.

Introduction

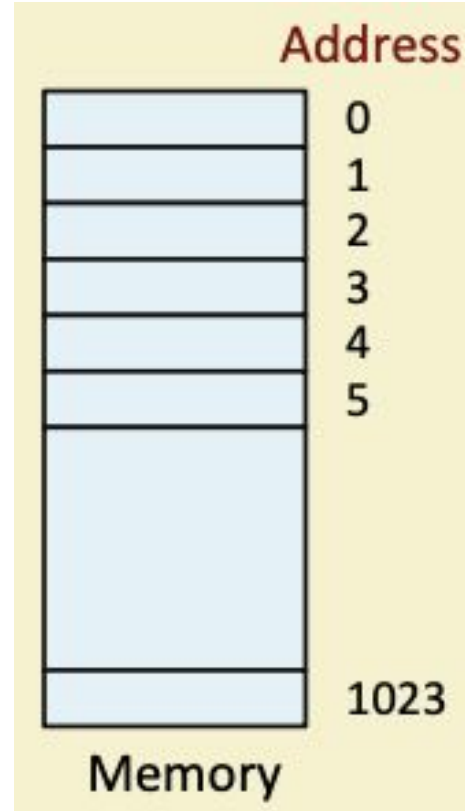
The basic mechanism through which an instruction gets executed shall be illustrated.

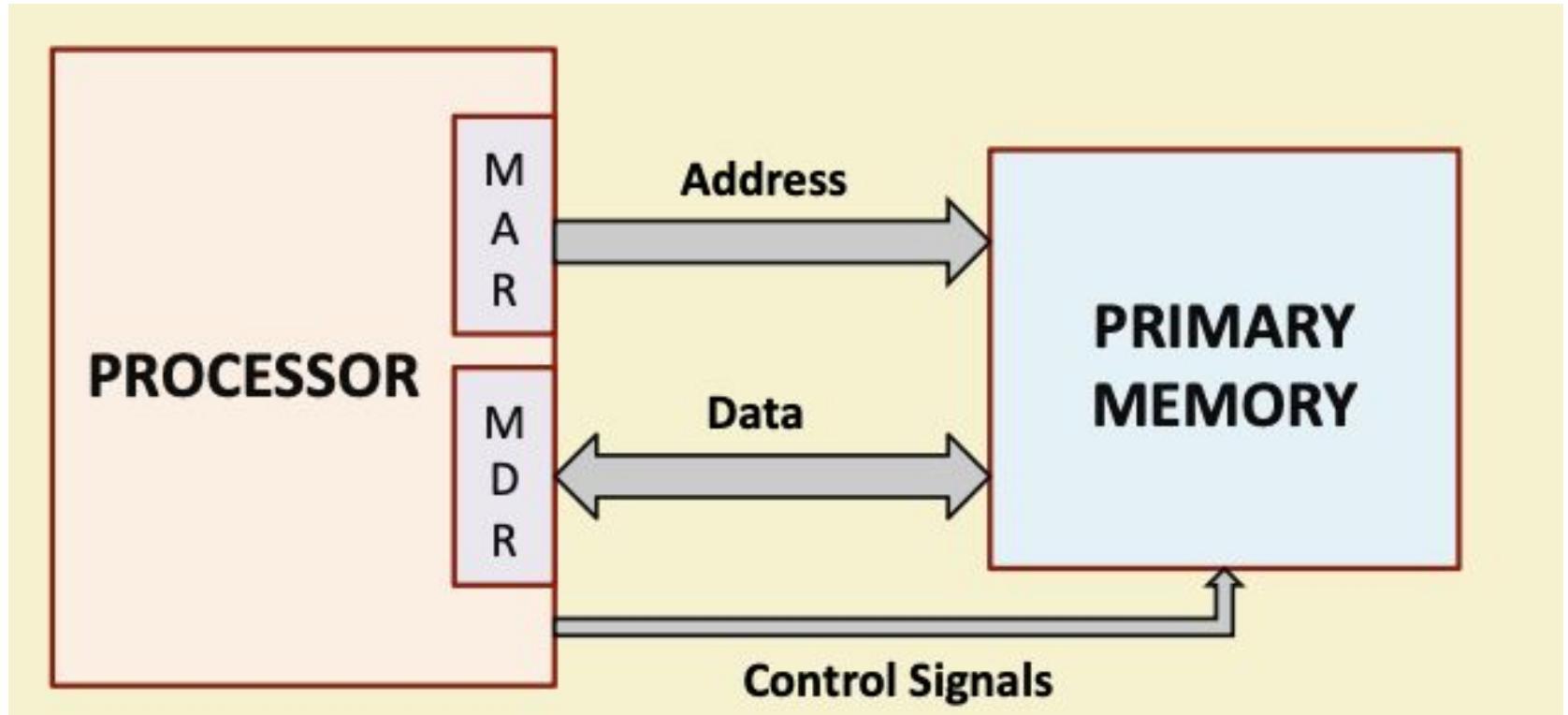
- May be recalled:
 - ALU contains a set of registers, some general-purpose and some special- purpose.
 - First we briefly explain the functions of the special-purpose registers before we look into some examples.



For Interfacing with the Primary Memory

- Two special-purpose registers are used:
 - Memory Address Register (MAR): Holds the address of the memory location to be accessed.
 - Memory Data Register (MDR): Holds the data that is being written into memory, or will receive the data being read out from memory.
- Memory considered as a linear array of storage locations (bytes or words) each with unique address.





- To read data from memory
 - a) Load the memory address into MAR.
 - b) Issue the control signal READ.
 - c) The data read from the memory is stored into MDR.
- To write data into memory
 - a) Load the memory address into MAR.
 - b) Load the data to be written into MDR.
 - c) Issue the control signal WRITE.



For Keeping Track of Program / Instructions

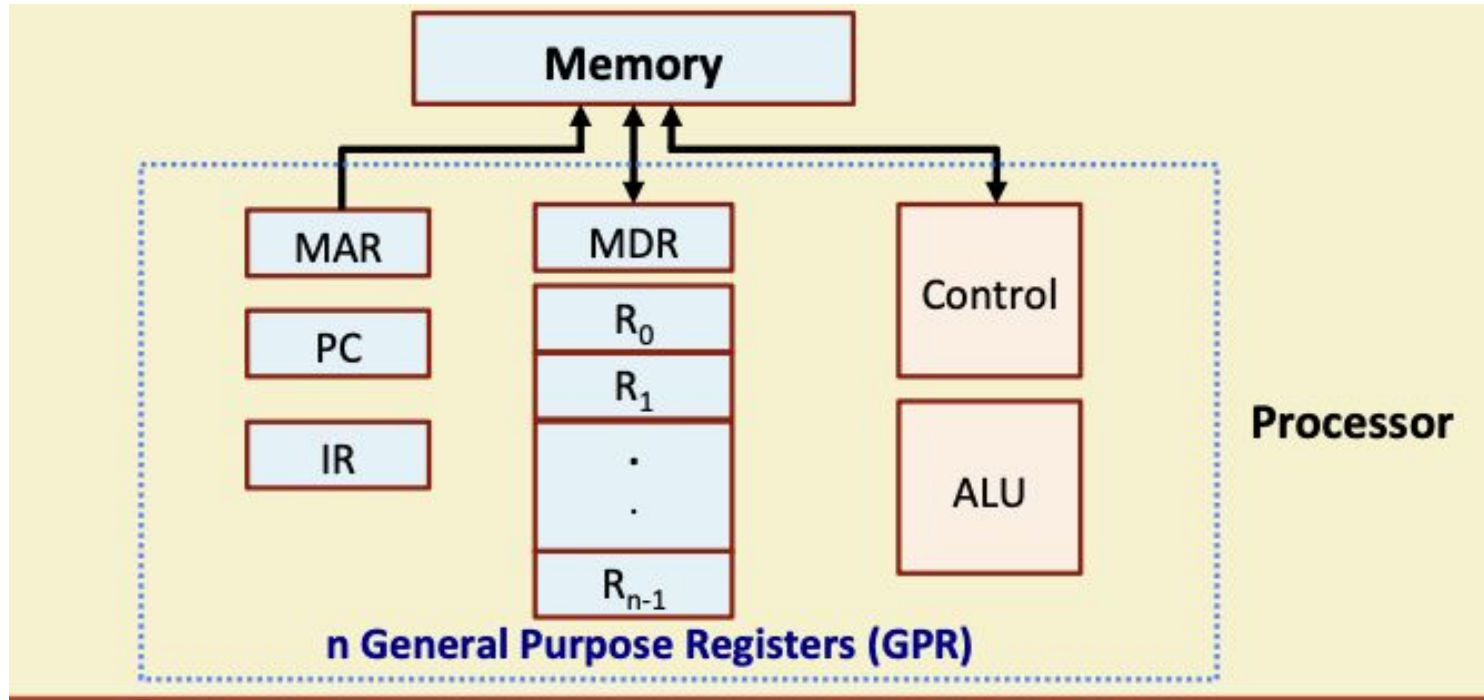
- Two special-purpose registers are used:
 - Program Counter (PC): Holds the memory address of the next instruction to be executed.
 - Automatically incremented to point to the next instruction when an instruction is being executed.
 - Instruction Register (IR): Temporarily holds an instruction that has been fetched from memory.
 - Need to be decoded to find out the instruction type.



Also contains information about the location of the data.

This is a property of Department of Computer Science, Faculty of Applied Science, University of Sri Jayewardenepura.

Architecture of the Example Processor



Example Instructions

- We shall illustrate the process of instruction execution with the help of the following two instructions:

a) ADD R1, LOCA

Add the contents of memory location LOCA (i.e. address of the memory location is LOCA) to the contents of register R1.

$R1 \leftarrow R1 + \text{Mem}[\text{LOCA}]$

b) ADD R1, R2

Add the contents of register R2 to the contents of register R1.

$R1 \leftarrow R1 + R2$



Execution of ADD R1,LOCA

Assume that the instruction is stored in memory location 1000, the initial value of R1 is 50, and LOCA is 5000.

- Before the instruction is executed, PC contains 1000.
- Content of PC is transferred to MAR.
- READ request is issued to memory unit.
- The instruction is fetched to MDR.
- Content of MDR is transferred to IR.
- PC is incremented to point to the next instruction
- The instruction is decoded by the control unit.

MAR $\leftarrow$ PC

MDR $\leftarrow$ Mem[MAR]

IR $\leftarrow$ MDR

PC $\leftarrow$ PC + 4

ADD R1

5000



Reference

1. Patterson, D. A., & Hennessy, J. L. (2013). Computer Organization and Design: The Hardware/Software Interface.
2. Stallings, W. (2018). Computer Organization and Architecture: Designing for Performance.
3. Tanenbaum, A. S., & Austin, T. (2012). Structured Computer Organization.
4. Murdocca, M. J., & Heuring, V. P. (2007). Computer Architecture and Organization: An Integrated Approach.



- LOCA (i.e. 5000) is transferred (from IR) to MAR.
- READ request is issued to memory unit.
- The data is fetched to MDR.
- The content of MDR is added to R1.

$MAR \leftarrow IR[Operand]$

$MDR \leftarrow Mem[MAR]$

$R1 \leftarrow R1 + MDR$

The steps being carried out are called micro-operations:

$MAR \leftarrow PC$

$MDR \leftarrow Mem[MAR]$

$IR \leftarrow MDR$

$PC \leftarrow PC + 4$

$MAR \leftarrow IR[Operand]$

$MDR \leftarrow Mem[MAR]$

$R1 \leftarrow R1 + MDR$



R1

125

Address	Content
1000	ADD R1, LOCA
1004	...

5000	75
------	----

LOCA

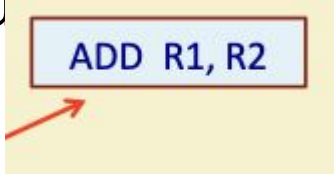
1. PC = 1000
2. MAR = 1000
3. PC = $PC + 4 = 1004$
4. MDR = ADD R1, LOCA
5. IR = ADD R1, LOCA
6. MAR = LOCA = 5000
7. MDR = 75
8. R1 = $R1 + MDR = 50 + 75 = 125$



Execution of ADD R1,R2

Assume that the instruction is stored in memory location 1500, the initial value of R1 is 50, and R2 is 200.

- Before the instruction is executed, PC contains 1500.
- Content of PC is transferred to MAR.
- READ request is issued to memory unit.
- The instruction is fetched to MDR.
- Content of MDR is transferred to IR.
- PC is incremented to point to the next instruction.
- The instruction is decoded by the control unit.
- R2 is added to R1.



ADD R1, R2

MAR $\leftarrow$ PC

MDR $\leftarrow$ Mem[MAR]

IR $\leftarrow$ MDR

PC $\leftarrow$ PC + 4

R1 $\leftarrow$ R1 + R2



R1 250

R2 200

Address	Instruction
1500	ADD R1, R2
1504	...

1. PC = 1500
2. MAR = 1500
3. PC = $PC + 4 = 1504$
4. MDR = ADD R1, R2
5. IR = ADD R1, R2
6. R1 = $R1 + R2 = 250$

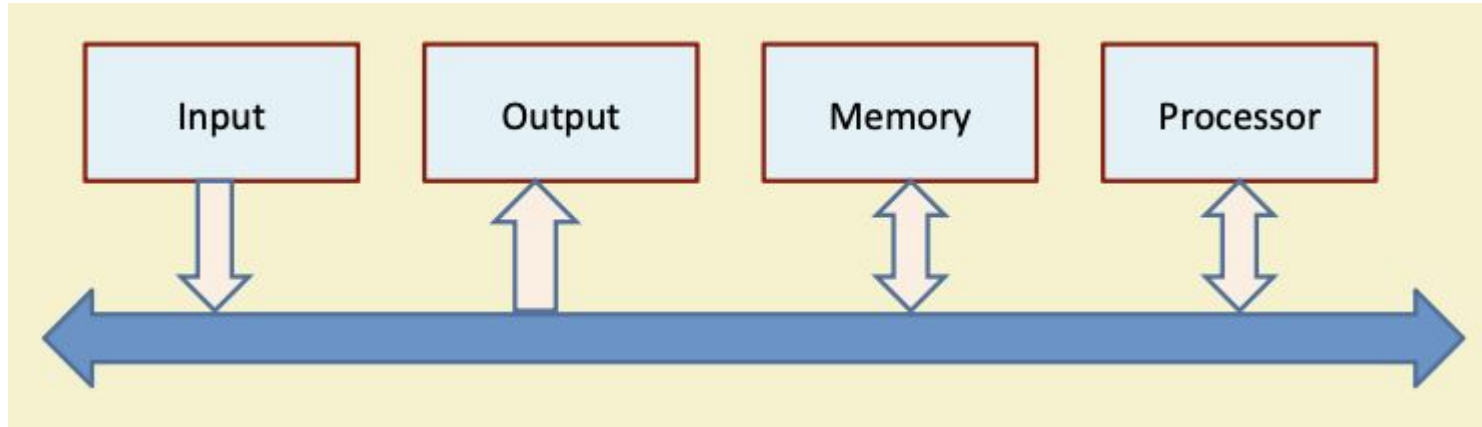


Bus Architecture

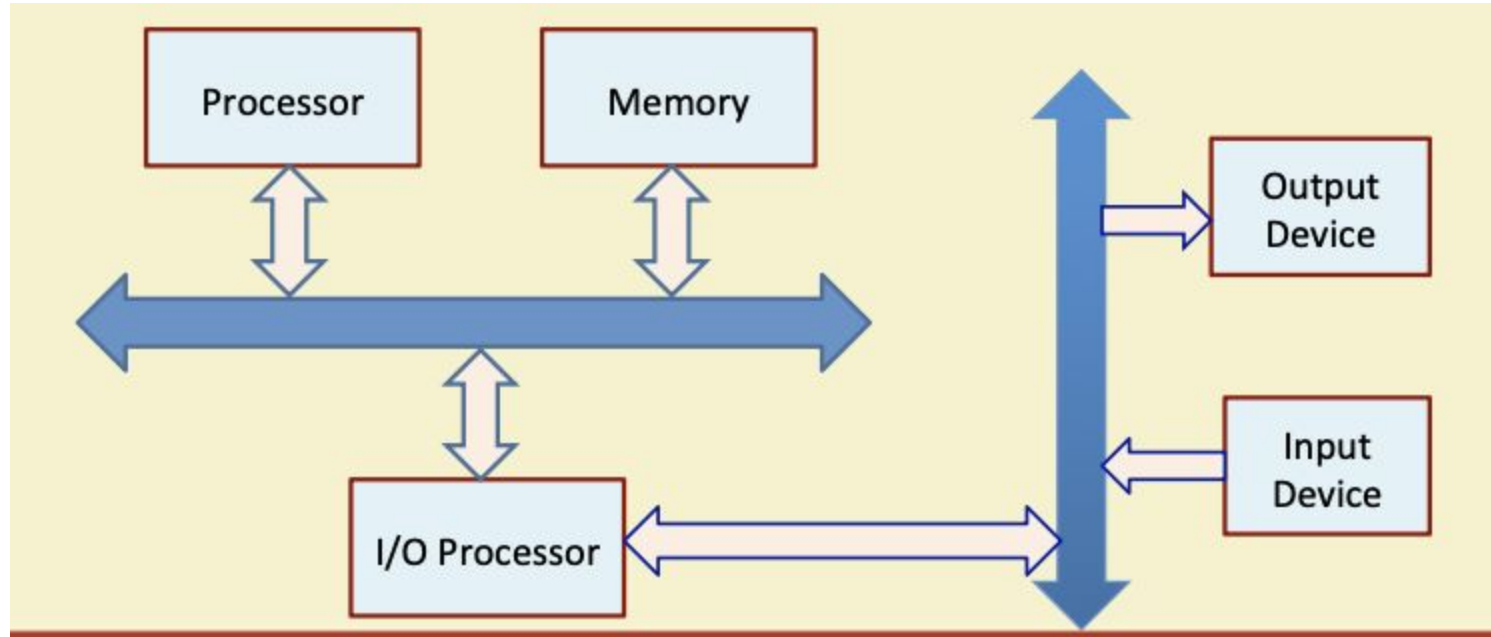
- The different functional modules must be connected in an organized manner to form an operational system.
- Bus refers to a group of lines that serves as a connecting path for several devices.
- The simplest way to connect the functional unit is to use the single bus architecture.
 - Only one data transfer allowed in one clock cycle.
 - For multi-bus architecture, parallelism in data transfer is allowed.



System-Level Single Bus Architecture



System-Level Two Bus Architecture



Single-Bus Architecture Inside the Processor

There is a single bus inside the processor.

- ALU and the registers are all connected via the single bus.
- This bus is internal to the processor and should not be confused with the external bus that connects the processor to the memory and I/O devices.

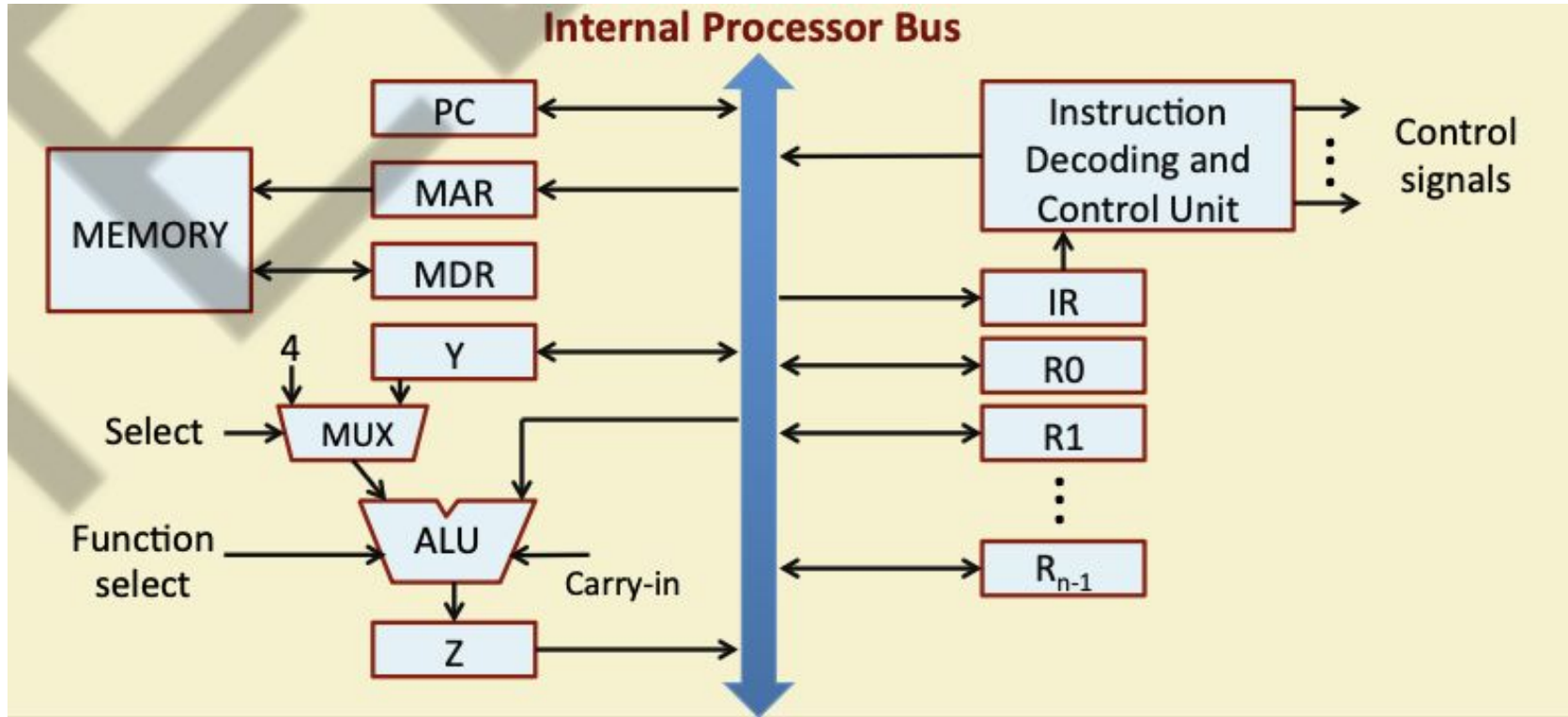


Single-Bus Architecture Inside the Processor

- A typical single-bus processor architecture is shown on the next slide.
 - Two temporary registers Y and Z are also included.
 - Register Y temporarily holds one of the operands of the ALU.
 - Register Z temporarily holds the result of the ALU operation.
 - The multiplexer selects a constant operand during execution of the micro- operation: $PC \leftarrow PC + 4$.



Internal Processor Bus



Multi-Bus Architecture

- Modern processors have multiple buses that connect the registers and other functional units.
 - Allows multiple data transfer micro-operations to be executed in the same clock cycle.
 - Results in overall faster instruction execution.
- Also advantageous to have multiple shorter buses rather than a single long bus.
 - Smaller parasitic capacitance, and hence smaller delay.



Thank
You!

